

Apellidos:.....; Nombre:.....

Para acondicionar una nave industrial se utiliza una máquina de refrigeración por absorción con un $COP=1.3$ (ver figura). Para producir el efecto frigorífico la máquina de refrigeración es alimentada con vapor saturado a 180°C generado en una caldera de propano (C_3H_8), retornado el vapor a la caldera como líquido saturado. La máquina frigorífica produce agua a 5°C que se envía a un evaporador en el que se enfría el agua que refrigera la nave de 15°C a 9°C .

Se desea conocer, por un lado, el consumo de propano (Nm^3/hr) de la caldera sabiendo que los humos se evacúan a 200°C , que el exceso de aire para la combustión completa es del 20% y que hay unas pérdidas del vapor generado del 1%, el porcentaje de agua de reposición respecto al caudal evaporado si dicha agua de reposición tiene una concentración de carbonatos del 2g/l y el fabricante aconseja que dichos niveles estén por debajo de 7g/l , y finalmente, es necesario seleccionar el intercambiador (evaporador) adecuado para nuestra aplicación de entre los que se adjuntan al final.

Datos sobre la caldera:

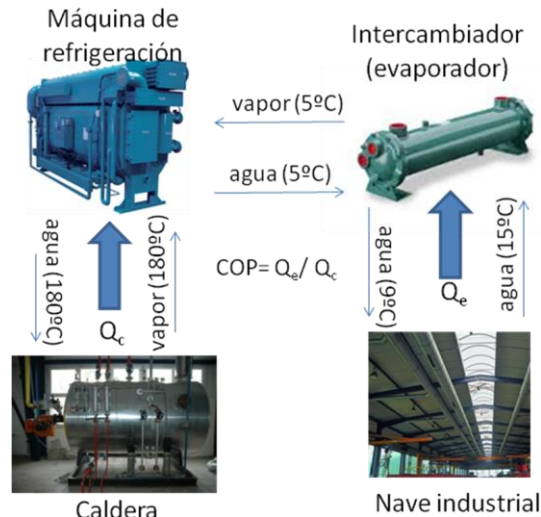
El aire ambiente se compone de 3.76Nm^3 de N_2 por cada Nm^3 de O_2 . El poder calorífico medio para los productos de la combustión es de $1.35\text{kJ}/\text{Nm}^3$. ($22.4\text{Nm}^3=1\text{kmol}$). PCI del propano $92\text{MJ}/\text{Nm}^3$.

Datos sobre la nave industrial:

La nave está situada en la planta baja de un edificio en España y consta de 3 muros exteriores, cada uno de ellos con un área de 250m^2 y con orientaciones norte, sur y este respectivamente. El otro muro de la nave por el cual se accede a ésta linda con otro espacio del mismo edificio.

- Temperatura de todo el edificio incluida la nave: 25°C .
- Humedad relativa en el interior de la nave: 50%.
- Composición de los muros: Ladrillo hueco (9cm de espesor), fibra de vidrio tipo I (5cm de espesor), enlucido de yeso (1.5cm de espesor).
- Temperatura seca exterior de diseño: 35°C
- Temperatura húmeda coincidente: 23°C
- Oscilación media diaria: 16°C
- Carga interna de la nave: 96kW
- Caudal de ventilación en la nave: $6\text{m}^3/\text{s}$
- Coeficiente de seguridad cálculo de la carga térmica: 10%.

Otros datos: $c_{p,\text{aire}}=1.02\text{kJ}/\text{kgK}$; $\rho_{\text{aire}}=1.2\text{kg}/\text{m}^3$; calor de cambio de fase del agua a 0°C : $h_{fg}=2501.3\text{kJ}/\text{kg}$.



Diferencia equivalente de temperatura para unas condiciones de $t_{\text{exterior}} = 35^{\circ}\text{C}$, $t_{\text{interior}} = 27^{\circ}\text{C}$, OMD= 11°C , mes de Julio, 40° latitud norte.

Δt_e	ORIENTACION				
Hora solar	Este	Oeste	Sur	Norte	Techo
6	2.8	3.9	2.2	0.5	5
7	2.8	3.9	2.2	0.5	4.4
8	3.3	3.3	1.1	0	3.3
9	4.4	3.3	1.1	0	3.9
10	7.8	3.3	1.1	0	4.4
11	11.1	3.3	1.7	0	6.1
12	13.3	3.3	2.2	0	8.9
13	13.9	3.9	4.4	0.5	12.2
14	13.3	4.4	6.7	1.1	15
15	11.1	5.5	8.3	1.7	17.2
16	10	6.7	8.9	2.2	19.4
17	8.9	9.4	10	2.8	21.2
18	7.8	11.1	10	2.8	21.7
19	7.8	13.9	8.3	2.8	21.1

Factor de corrección de la Δt_e					
Corrección	OMD en 24h				
$t_{se,\text{diseño}} - t_{sL}$	8	10	12	14	16
0	-6.6	-7.6	-8.5	-9.4	-10.3
4	-2.7	-3.6	-4.6	-5.5	-6.4
8	1.2	0.3	-0.7	-1.6	-2.5
10	3.1	2.2	1.2	0.3	-0.6



Placas – welded:

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot-side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	83 2.33	133 1.8	NA	238 1.67	176 1.96	134 1.95	246 0.934	191 1.01	170 1.96
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	129 2.1	503 0.733	NA	1262 0.43	819 0.54	774 0.61	1384 0.494	850 0.634	440 0.52
	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	210 1.53	1068 0.42	NA	9100 0.147	4268 0.26	2067 0.26	5465 0.18	1518 0.274	600 0.55
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	141 1.82	1060 0.37	NA	3570 0.21	3763 0.194	1075 0.367	3652 0.214	1366 0.297	600 0.55
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	131 1.9	298 0.87	NA	484 0.52	413 0.584	292 0.824	1626 0.287	403 0.54	300 0.87
	Boiling Water	U (W/m ² K) C (€/(W/K))	220 1.51	1100 0.42	NA	9000 0.15	4500 0.25	2000 0.26	5500 0.18	1500 0.28	600 0.55
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	160 1.65	900 0.39	NA	5000 0.19	3500 0.20	1200 0.37	4000 0.26	1000 0.37	500 0.53
	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	65 3.84	141 1.33	NA	230 1.42	167 1.74	141 2.0	246 1.06	190 0.92	170 1.75
100 000	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	137 1.38	534 0.462	NA	1354 0.303	940 0.34	776 0.41	1342 0.238	834 0.307	440 0.51
	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	NA	NA	NA	NA	NA	NA	NA	NA	NA
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	223 1.37	1112 0.28	NA	9420 0.062	4858 0.108	2106 0.13	5071 0.085	1518 0.186	600 0.40
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	157 1.71	722 0.36	NA	4140 0.118	3794 0.101	1277 0.19	3712 0.108	1367 0.161	600 0.40
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	133 1.67	355 0.592	NA	530 0.42	415 0.538	302 0.515	1617 0.158	452 0.389	300 0.52
	Boiling Water	U (W/m ² K) C (€/(W/K))	220 1.37	1100 0.28	NA	9000 0.063	4500 0.13	2000 0.15	5500 0.09	1500 0.17	600 0.40
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	160 1.71	900 0.32	NA	5000 0.12	3500 0.14	1200 0.19	4000 0.13	1000 0.30	500 0.43
	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 1.11	93 0.76	120 1.06	102 0.73	99 0.74	63 0.99	107 0.71	100 0.73	86 0.80
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	93 0.76	300 0.37	350 0.62	500 0.28	375 0.33	120 0.63	530 0.28	388 0.32	240 0.42
30 000	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	120 1.06	350 0.62	400 0.94	600 0.40	450 0.53	200 0.73	600 0.40	400 0.62	300 0.62
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	105 0.71	484 0.29	600 0.40	938 0.23	714 0.25	142 0.56	1607 0.19	764 0.24	345 0.34
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.38	130 0.59	818 0.24	524 0.28	286 0.38
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	68 0.94	138 0.57	200 0.73	161 0.52	153 0.53	82 0.83	173 0.50	155 0.53	124 0.60
	Boiling Water	U (W/m ² K) C (€/(W/K))	105 0.71	467 0.29	550 0.49	875 0.23	677 0.25	140 0.56	1432 0.20	722 0.25	336 0.35
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.28	130 0.59	818 0.24	524 0.28	286 0.38
	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 0.95	93 0.58	120 0.93	102 0.54	99 0.55	63 0.83	107 0.52	100 0.55	86 0.62
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	93 0.58	300 0.23	350 0.35	429 0.18	375 0.20	120 0.47	530 0.16	388 0.19	240 0.27
	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	120 0.93	350 0.35	400 0.58	600 0.24	450 0.28	200 0.64	600 0.24	400 0.32	300 0.39
100 000	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	105 0.52	484 0.17	600 0.24	938 0.116	714 0.134	142 0.41	1607 0.086	764 0.129	345 0.21
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.55	375 0.20	450 0.28	609 0.145	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	68 0.77	138 0.42	200 0.64	161 0.37	153 0.38	82 0.65	173 0.35	155 0.38	124 0.45
	Boiling Water	U (W/m ² K) C (€/(W/K))	105 0.52	467 0.168	550 0.26	875 0.121	677 0.137	140 0.41	1432 0.091	722 0.133	336 0.22
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.55	375 0.20	450 0.28	600 0.146	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 0.95	93 0.58	120 0.93	102 0.54	99 0.55	63 0.83	107 0.52	100 0.55	86 0.62
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	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	120 0.93	350 0.35	400 0.58	600 0.24	450 0.28	200 0.64	600 0.24	400 0.32	300 0.39
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	105 0.52	484 0.17	600 0.24	938 0.116	714 0.134	142 0.41	1607 0.086	764 0.129	345 0.21

Carcasa y tubos:

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 1.11	93 0.76	120 1.06	102 0.73	99 0.74	63 0.99	107 0.71	100 0.73	86 0.80
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	93 0.76	300 0.37	350 0.62	500 0.28	375 0.33	120 0.63	530 0.28	388 0.32	240 0.42
	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	120 1.06	350 0.62	400 0.94	600 0.40	450 0.53	200 0.73	600 0.40	400 0.62	300 0.62
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	105 0.71	484 0.29	600 0.40	938 0.23	714 0.25	142 0.56	1607 0.19	764 0.24	345 0.34
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.38	130 0.59	818 0.24	524 0.28	286 0.38
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	68 0.94	138 0.57	200 0.73	161 0.52	153 0.53	82 0.83	173 0.50	155 0.53	124 0.60
	Boiling Water	U (W/m ² K) C (€/(W/K))	105 0.71	467 0.29	550 0.49	875 0.23	677 0.25	140 0.56	1432 0.20	722 0.25	336 0.35
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.28	130 0.59	818 0.24	524 0.28	286 0.38
	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 0.95	93 0.58	120 0.93	102 0.54	99 0.55	63 0.83	107 0.52	100 0.55	86 0.62
100 000	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	93 0.58	300 0.23	350 0.35	429 0.18	375 0.20	120 0.47	530 0.16	388 0.19	240 0.27
	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	120 0.93	350 0.35	400 0.58	600 0.24	450 0.28	200 0.64	600 0.24	400 0.32	300 0.39
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	105 0.52	484 0.17	600 0.24	938 0.116	714 0.134	142 0.41	1607 0.086	764 0.129	345 0.21
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.55	375 0.20	450 0.28	609 0.145	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	68 0.77	138 0.42	200 0.64	161 0.37	153 0.38	82 0.65	173 0.35	155 0.38	124 0.45
	Boiling Water	U (W/m ² K) C (€/(W/K))	105 0.52	467 0.168	550 0.26	875 0.121	677 0.137	140 0.41	1432 0.091	722 0.133	336 0.22
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.55	375 0.20	450 0.28	600 0.146	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 0.95	93 0.58	120 0.93	102 0.54	99 0.55	63 0.83	107 0.52	100 0.55	86 0.62
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Cálculo de la carga térmica de refrigeración.

Datos:

Propiedades del aire

$$c_p = 1020 \frac{\text{J}}{\text{kg} \cdot \text{C}} \quad \rho = 1.2 \frac{\text{kg}}{\text{m}^3} \quad h_{fg} = 2501.3 \cdot 10^3 \frac{\text{J}}{\text{kg}}$$

Características del muro

$$e_l = 0.09 \text{ m} \quad e_f = 0.05 \text{ m} \quad e_y = 0.015 \text{ m} \quad A_M = 250 \text{ m}^2$$

Condiciones en el local:

$$t_{sL} = 25 \text{ C} \quad \phi_L = 50 \%$$

Condiciones en el exterior

$$t_{se} = 35 \text{ C} \quad t_{hc} = 23 \text{ C}$$

$$\text{OMD} = 16 \text{ C}$$

$$t_{se} - t_{sL} = 10 \cdot \text{C}$$

Carga interna de la nave:

$$Q_{int} = 96 \text{ kW}$$

Caudal de ventilación:

$$V_{vent} = 6 \frac{\text{m}^3}{\text{s}}$$

Carga Transmisión:

De la norma básica NBE_CT_79

Del documento básico de ahorro de energía:

$$R_l = 0.18 \frac{\text{m}^2 \cdot \text{C}}{\text{W}}$$

$$R_e = 0.04 \frac{\text{m}^2 \cdot \text{C}}{\text{W}}$$

$$k_f = 0.044 \frac{\text{W}}{\text{m} \cdot \text{C}}$$

$$R_i = 0.13 \frac{\text{m}^2 \cdot \text{C}}{\text{W}}$$

$$k_y = 0.30 \frac{\text{W}}{\text{m} \cdot \text{C}}$$

$$U_M = \frac{1}{R_l + R_i + \frac{e_f}{k_f} + \frac{e_y}{k_y} + R_e} = 0.651 \cdot \frac{\text{W}}{\text{m}^2 \cdot \text{C}}$$



Diferencia de temperaturas equivalente:

$$\Delta t_{\text{etablaN}} := 1.7^\circ\text{C}$$

$$\Delta t_{\text{etablaS}} := 8.3^\circ\text{C}$$

$$\text{Corr} := -0.6^\circ\text{C}$$

$$\Delta t_{\text{eN}} := \Delta t_{\text{etablaN}} + \text{Corr} = 1.1 \cdot ^\circ\text{C}$$

$$\Delta t_{\text{eS}} := \Delta t_{\text{etablaS}} + \text{Corr} = 7.7 \cdot ^\circ\text{C}$$

$$\Delta t_{\text{eE}} := \Delta t_{\text{etablaE}} + \text{Corr} = 10.5 \cdot ^\circ\text{C}$$

$$Q_M := U_M \cdot A_M \cdot \Delta t_{\text{eN}} + U_M \cdot A_M \cdot \Delta t_{\text{eS}} + U_M \cdot A_M \cdot \Delta t_{\text{eE}} = 3.141 \cdot \text{kW}$$

Ventilación:

$$Q_{\text{ventsen}} := V_{\text{vent}} \cdot \rho \cdot c_p \cdot (t_{\text{se}} - t_{\text{SL}}) = 7.344 \times 10^4 \cdot \text{W}$$

Del diagrama psicrométrico

$$w_e := 0.013$$

$$w_L := 0.010$$

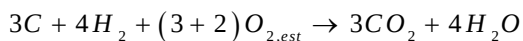
$$Q_{\text{ventlat}} := V_{\text{vent}} \cdot \rho \cdot h_{fg} \cdot (w_e - w_L) = 5.403 \times 10^4 \cdot \text{W}$$

$$Q_{\text{vent}} := Q_{\text{ventsen}} + Q_{\text{ventlat}} = 127.468 \cdot \text{kW}$$

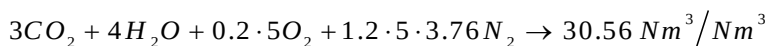
$$Q_T := 1.1(Q_M + Q_{\text{int}} + Q_{\text{vent}}) = 249.269 \cdot \text{kW}$$

Cálculo de la caldera:

Oxígeno estequiométrico



Volumen de humos



Pérdidas por la chimenea

$$Q_{\text{humos}} / m_c = 30.56 \frac{\text{Nm}^3}{\text{Nm}^3} 1.35 \frac{\text{kJ}}{\text{Nm}^3 \cdot ^\circ\text{C}} (200 - 35) ^\circ\text{C} = 6.81 \frac{\text{MJ}}{\text{Nm}^3}$$

$$\frac{Q_{\text{humos}} / m_c}{PCI} = 7.4\%$$

Rendimiento

$$\eta = \frac{Q_c}{m_c PCI} = 1 - \frac{Q_{\text{humos}}}{m_c PCI} - \frac{Q_{\text{vapor}}}{m_c PCI} = 1 - \frac{Q_{\text{humos}}}{m_c PCI} - \frac{0.01 \frac{Q_c}{h_{fg,180^\circ\text{C}}} h_{fg,180^\circ\text{C}}}{m_c PCI} = 1 - \frac{Q_{\text{humos}} / m_c}{PCI} - 0.01\eta$$



$$\frac{1 - (Q_{humos}/m_c)/PCI}{1.01} = \frac{1 - 0.074}{1.01} = \frac{0.926}{1.01} = 91.68\%$$

$$Q_c = Q_e/COP = 249.269/1.3 \cdot \text{kW} = 191.74 \text{ kW}$$

$$m_c = \frac{Q_c}{\eta PCI} = \frac{192.1}{0.9168 \cdot 92000} = 8.18 \text{ Nm}^3/\text{hr}$$

$$\frac{R}{m_v} = \frac{c_1}{c_1 - c_0} \frac{F_v}{m_v} = \frac{7}{7 - 2} 0.01 = 1.4\%$$

Selección del tipo de intercambiador de calor (Evaporador)

$$\Delta T_{lm} = \frac{(T_{a,e} - T_e) - (T_{a,s} - T_e)}{\ln \frac{T_{a,e} - T_e}{T_{a,s} - T_e}} = \frac{(15 - 5) - (9 - 5)}{\ln \frac{15 - 5}{9 - 5}} = 6.55^\circ \text{C}$$

$$UA = Q_e / \Delta T_{lm} = 249.27 \cdot 10^3 / 6.55 = 38067 \frac{\text{W}}{\text{K}}$$

Interpolando en ambas tablas para el UA obtenido (fluidos de trabajo dentro de recuadros rojos):

Intercambiador de placas-welded:

Coste total: 4810 libras esterlinas

$$\text{Área total de intercambio de calor: } A_{pw} = UA / U_{pw} = 38067 / 9000 = 4.23 \text{ m}^2$$

Intercambiador de carcasa y tubos:

Coste total: 7711 libras esterlinas

$$\text{Área total de intercambio de calor: } A_{ct} = UA / U_{ct} = 38067 / 875 = 43.51 \text{ m}^2$$

Por tanto, en este caso sería recomendable elegir un intercambiador de placas-welded.